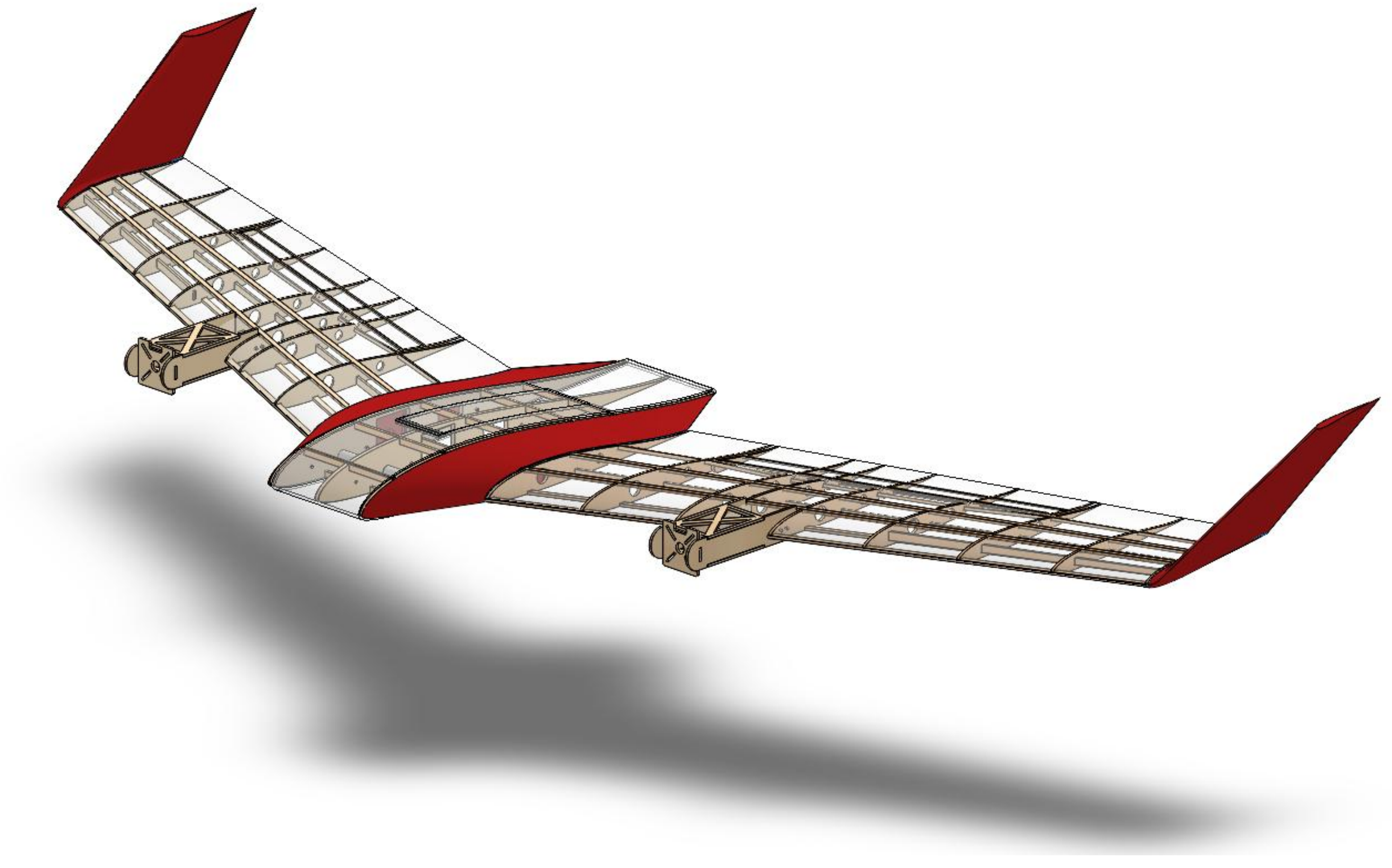
Project Management Plan (PMP)

for the

VTOL Flying Wing Tail-Sitter Project



WBS 1.0 VTOL Flying Wing Tail-Sitter

The objective of the VTOL Flying Wing Tail-Sitter project is to execute the conceptual design, sizing, aerodynamic analysis, propulsion design, structural design, and flight-control development required to produce a validated Phase I design package for a flying-wing tail-sitter unmanned aerial vehicle. The project supports the Aerospace Engineering Department's Third-Year Project requirement and establishes the technical baseline (sizing, structure, propulsion, and control) from which a physical prototype can subsequently be manufactured and flight-tested.

WBS 1.01 Literature Review & Requirements Definition Scope includes the technical research and requirements-setting effort that anchors all downstream design decisions, spanning aerodynamic configuration studies, structural load literature, and control-strategy research for tail-sitter VTOL aircraft.

WBS 1.01 Literature Review & Requirements Definition

WBS 1.01.01 Aerodynamic & Configuration Literature Review: Survey of VTOL configurations (standard fixed-wing VTOL, tiltrotor, tiltwing, and tail-sitter), review of precedent tail-sitter aircraft and case studies, aero-propulsive slipstream effects, immersion ratio and control-effectiveness research, and a weighted trade-off analysis leading to configuration selection and conceptual design methodology, including reflex-airfoil tailless-stability theory.

WBS 1.01.02 Mechanical & Structural Literature Review: Review of tail-sitter structural load environments, aerodynamic load sharing between wing and fuselage, spanwise load distribution, propeller-induced load distortion during transition, structural load paths and critical joints, impact/inertial loading from pivot landing and take-off, composite weight-reduction methods, spar geometry theory, ply-optimization algorithms, and mechanical joint and hybrid CFRP-to-metal joint technologies.

WBS 1.01.03 Control Systems Literature Review: Survey of tail-sitter control-wise modeling and control strategies, classical and modern control techniques, optimal and robust control approaches, transition-trajectory planning, model predictive control, adaptive/hybrid architectures, and state-estimation methods including the Extended Kalman Filter.

WBS 1.01.04 Mission & Requirements Definition: Definition of the tail-sitter mission profile and top-level performance requirements that anchor the sizing, aerodynamic, propulsion, structural, and control work packages that follow.

WBS 1.02 Conceptual Sizing & Design Point Selection

WBS 1.02.01 Weight Estimation: Weight decomposition, empty-weight model development, cruise and hover battery weight estimation, iterative convergence solution, weight-fraction sanity checks, and sensitivity analysis to cruise endurance.

WBS 1.02.02 Matching Plot & Design Point Selection: Mathematical formulation of rotorcraft and fixed-wing performance constraints, definition of atmospheric, rotorcraft, and fixed-wing input parameters, construction of the matching plot and feasible regions, and selection of the final rotorcraft, fixed-wing, and combined design points.

WBS 1.02.03 Airfoil Selection: Definition of operating lift coefficient and Reynolds-number range, establishment of airfoil selection criteria, XFLR5/XFOIL analysis setup, screening of candidate airfoils (MH45, MH60, PW51, PW75, S5010) at multiple Ncrit values, robustness checks, and final airfoil selection and justification.

WBS 1.02.04 Wing Aerodynamic Sizing: Derivation of wing reference area from wing loading, two-panel trapezoidal wing geometry definition (inner fuselage panel and outer wing panel), area-matching iteration, and longitudinal stability / static-margin verification.

WBS 1.03 Aerodynamic Analysis & Wing Design Verification

WBS 1.03.01 XFLR5 Wing Analysis. Iteration 1: Geometric representation of the wing, component placement and mass-distribution / centre-of-gravity definition, panel-mesh setup, pressure-coefficient distribution, and generation of lift, pitching-moment, aerodynamic-efficiency, and directional/lateral stability-derivative curves for the first design iteration.

WBS 1.03.02 Longitudinal & Lateral Stability Analysis. Iteration 1: Root-locus and time-response stability analysis for the first design iteration, covering the short-period and phugoid longitudinal modes and the roll, spiral, and Dutch-roll lateral modes.

WBS 1.03.03 XFLR5 Wing Analysis. Iteration 2: Revised geometric representation, updated mass distribution and centre-of-gravity placement, panel-mesh setup, pressure-coefficient distribution, and aerodynamic characteristic curves reflecting design refinements identified in Iteration 1.

WBS 1.03.04 Stability Analysis. Iteration 2 & Comparison: Root-locus and time-response stability analysis for the refined geometry, and a comparative summary of longitudinal and lateral stability characteristics between Iteration 1 and Iteration 2.

WBS 1.04 Propulsion System Design

WBS 1.04.01 Propulsion Sizing Requirements: Power-requirement derivation from the matching plot, thrust-requirement calculation, and disk-loading and propeller-diameter checks that bound the propulsion trade space.

WBS 1.04.02 Component Database & Candidate Screening: Compilation of motor, battery, and propeller databases, battery cell-count selection, RPM/KV search-range definition, and identification of pre-verification candidate motor/battery/propeller combinations.

WBS 1.04.03 Hover & Cruise Verification: Definition of hover verification criteria and execution of hover verification results, hover-screening of passed candidates, fixed-wing cruise verification criteria and results, and cruise-rejection screening.

WBS 1.04.04 Final Propulsion Selection & Documentation: Compilation of the passed complete candidate table, battery-capacity check, final propulsion-system selection and justification, and documentation of the selected motor.

WBS 1.05 Computational Fluid Dynamics (CFD)

WBS 1.05.01 Geometry Simplification: Simplification of the stationary airframe geometry and the rotating propeller geometry for CFD simulation, including comparison of the modeled propeller against the actual propeller geometry.

WBS 1.05.02 Domain & Mesh Generation: Computational domain identification and generation of the simulation mesh.

WBS 1.05.03 Simulation Setup & Results: CFD solver setup and execution, and analysis/reporting of the resulting flow-field data for the wing and propeller.

WBS 1.06 Mechanical Design & Manufacturing

WBS 1.06.01 Fuselage Structural Design: Fuselage (inner wing panel) geometry definition, rib configuration, stringer configuration, and anti-torsion spar design.

WBS 1.06.02 Wing Connector & Joint Design: Design and topology optimization of the wing connectors, spar-interface design (front and rear spar sockets), and connector-to-rib fastening design.

WBS 1.06.03 Wing Structural Design: Material and manufacturing methodology comparison, mass comparison and weight-budget implications, wing structural-component definition, rib-spacing determination (problem definition, governing equations, quantitative results), and front and rear spar sizing including load definitions, candidate evaluation, sizing-tool inputs/results, and analytical shear and moment diagrams.

WBS 1.06.04 Motor Mount Design & FEA: Motor-mount design and finite element analysis to validate structural adequacy under operating loads.

WBS 1.06.05 Material Selection & 3D-Printing Parameters: Material selection for structural and printed components, and definition of 3D-printing (FDM) parameters for the wing connectors, including slicer settings, infill, layer height, and support configuration.

WBS 1.06.06 Manufacturing, Assembly & Bill of Materials: Execution of the manufacturing sequence, internal layout and component-access planning, production of the assembly drawing, weight-budget tracking against target, and compilation of the bill of materials.

WBS 1.07 Flight Dynamics & Control System Design

WBS 1.07.01 Equations of Motion & Coordinate Transformations: Derivation of conservation laws, linear- and angular-momentum equations, modeling assumptions, reference-frame definitions, rotation matrices via Euler angles, gravitational force components, and kinematic equations.

WBS 1.07.02 Longitudinal Dynamics & Linear Control Model: Three-degree-of-freedom longitudinal formulation and physical interpretation, linearization methodology, stability-derivative computation, perturbed equations of motion, state-space representation, and definition of operating points and flight regimes.

WBS 1.07.03 Aerodynamic Modeling (Propeller Slipstream Effects): Modeling of downstream slipstream speed, total velocity over the propulsive section, effective angle of attack, wing-region decomposition, propulsive area, and the resulting lift, pitching-moment, and drag coefficient models, including conversion of coefficients to forces and moments.

WBS 1.07.04 Trim Analysis: Horizontal flight trim calculation, including flight condition and dynamic pressure, trim elevator deflection from moment balance, trim angle of attack from lift balance, trim thrust from drag balance, and sample numerical results.

WBS 1.07.05 3DOF Simulation & Controller Design: Construction of the simulation model, induced-velocity and aerodynamic (lift, moment, drag) model implementation, hover-controller design including X-position and altitude controllers, and simulation results covering position-tracking response, altitude-tracking response, and motion visualization.

WBS 1.08 Project Management, Integration & Reporting

WBS 1.08.01 Team Coordination & Scheduling: Task allocation and milestone scheduling across the eight-member project team, and coordination between the aerodynamics, propulsion, structures, and control sub-teams.

WBS 1.08.02 Design Integration & Internal Reviews: Integration of sizing, aerodynamic, propulsion, structural, and control outputs into a consistent design baseline, and conduct of internal design reviews to reconcile cross-discipline dependencies.

WBS 1.08.03 Documentation & Report Compilation: Compilation of the literature review, technical chapters, figures, and references into the Phase I report, and formatting for submission.

WBS 1.08.04 Submission & Advisor Review: Submission of the Phase I report to the project advisor and incorporation of resulting feedback into the design baseline.

VTOL Tail-Sitter Preliminary WBS Structure

1.0 VTOL Flying Wing Tail-Sitter

1.01 Literature Review & Requirements	1.02 Conceptual Sizing & Design Point	1.03 Aerodynamic Analysis	1.04 Propulsion System Design	1.05 CFD	1.06 Mechanical Design & Mfg.	1.07 Flight Dynamics & Control	1.08 Project Mgmt. & Reporting
1.01.01 Aero/Config. Lit. Review	1.02.01 Weight Estimation	1.03.01 XFLR5 Analysis Iter. 1	1.04.01 Sizing Requirements	1.05.01 Geometry Simplification	1.06.01 Fuselage Structural Design	1.07.01 Equations of Motion	1.08.01 Team Coordination
1.01.02 Mechanical/Struct. Lit. Review	1.02.02 Matching Plot & Design Pt.	1.03.02 Stability Iter. 1	1.04.02 Component DB & Screening	1.05.02 Domain & Mesh	1.06.02 Wing Connector & Joint Design	1.07.02 Longitudinal Dynamics	1.08.02 Design Integration & Reviews
1.01.03 Control Lit. Review	1.02.03 Airfoil Selection	1.03.03 XFLR5 Analysis Iter. 2	1.04.03 Hover & Cruise Verification	1.05.03 Simulation & Results	1.06.03 Wing Structural Design	1.07.03 Aerodynamic Modeling	1.08.03 Documentation & Report
1.01.04 Mission & Requirements	1.02.04 Wing Aerodynamic Sizing	1.03.04 Stability Iter. 2 & Compare	1.04.04 Final Propulsion Selection		1.06.04 Motor Mount Design & FEA	1.07.04 Trim Analysis	1.08.04 Submission & Advisor Review
					1.06.05 Material Selection & 3D-Print Params.	1.07.05 3DOF Sim. & Controller Design	
					1.06.06 Manufacturing, Assembly & BOM		

2. Project Schedule

The schedule below sequences all WBS 1.01–1.08 work packages on a generic project-week axis (Week 0 = kickoff). Durations and dependencies are derived directly from the logical build order in the WBS Dictionary: sizing cannot start before requirements are set, structural design cannot start before aerodynamic loads and propulsion mass are known, and so on. Map Week 0 to your actual project start date to convert this into a calendar schedule.

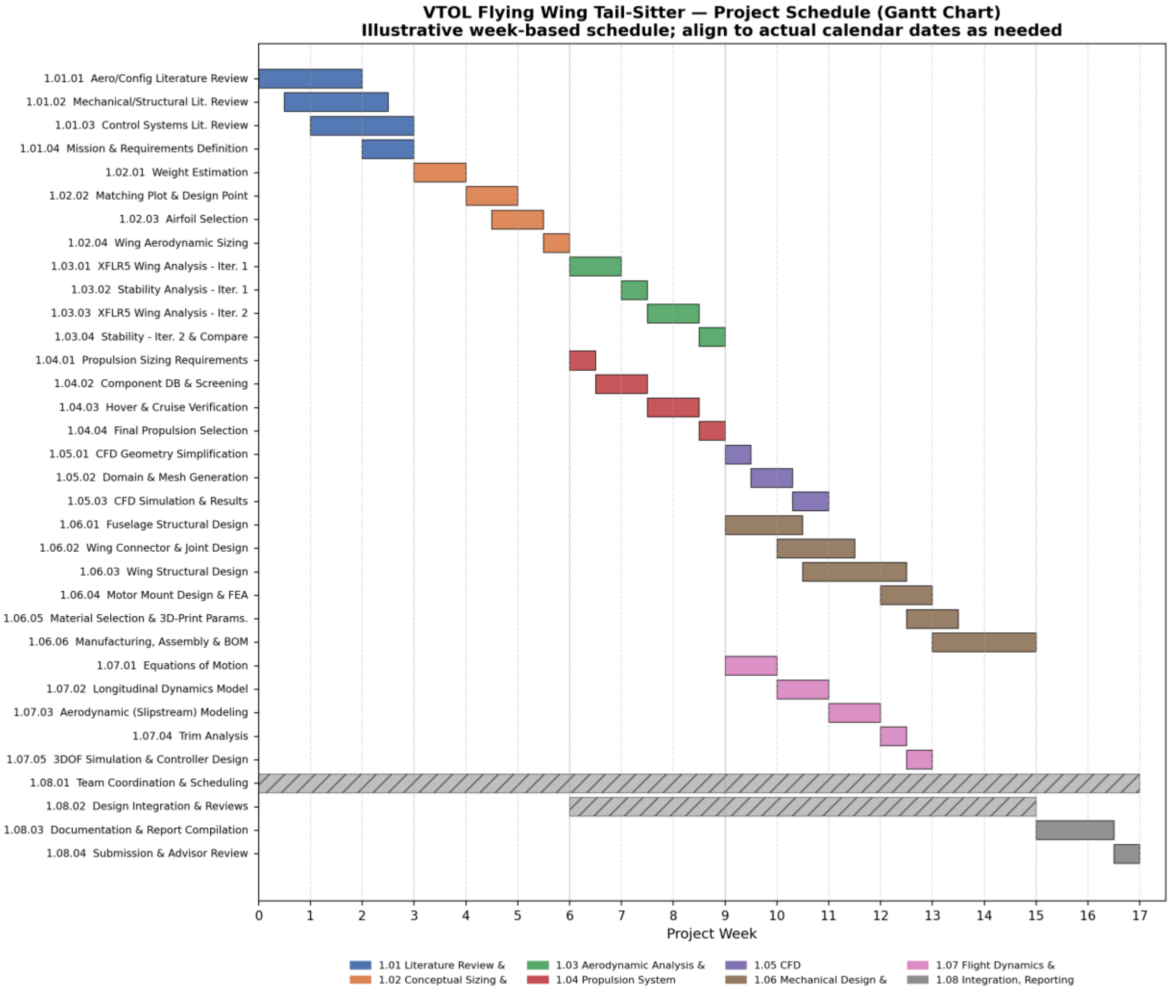


Figure 1: Gantt chart, WBS-level sub-tasks colour-coded by phase

2.1 Phase Summary

WBS	Phase	Duration (wk)	Start (wk)	Finish (wk)
1.01	Literature Review & Requirements	3	0	3
1.02	Conceptual Sizing & Design Point	3	3	6
1.03	Aerodynamic Analysis & Wing Verification	3	6	9
1.04	Propulsion System Design	3	6	9
1.05	CFD	2	9	11
1.06	Mechanical Design & Manufacturing	6	9	15
1.07	Flight Dynamics & Control	4	9	13
1.08	Integration, Reporting & Submission	2	15	17

Total planned duration: 17 weeks from kickoff to Phase I report submission.

3. Critical Path Analysis

A forward and backward pass (Critical Path Method) was run over the eight WBS phases to identify which activities have zero float meaning any delay to them delays the whole project and which carry schedule slack. ES/EF are the earliest a phase can start/finish; LS/LF are the latest it can start/finish without pushing the finish date; Float is the difference between them.

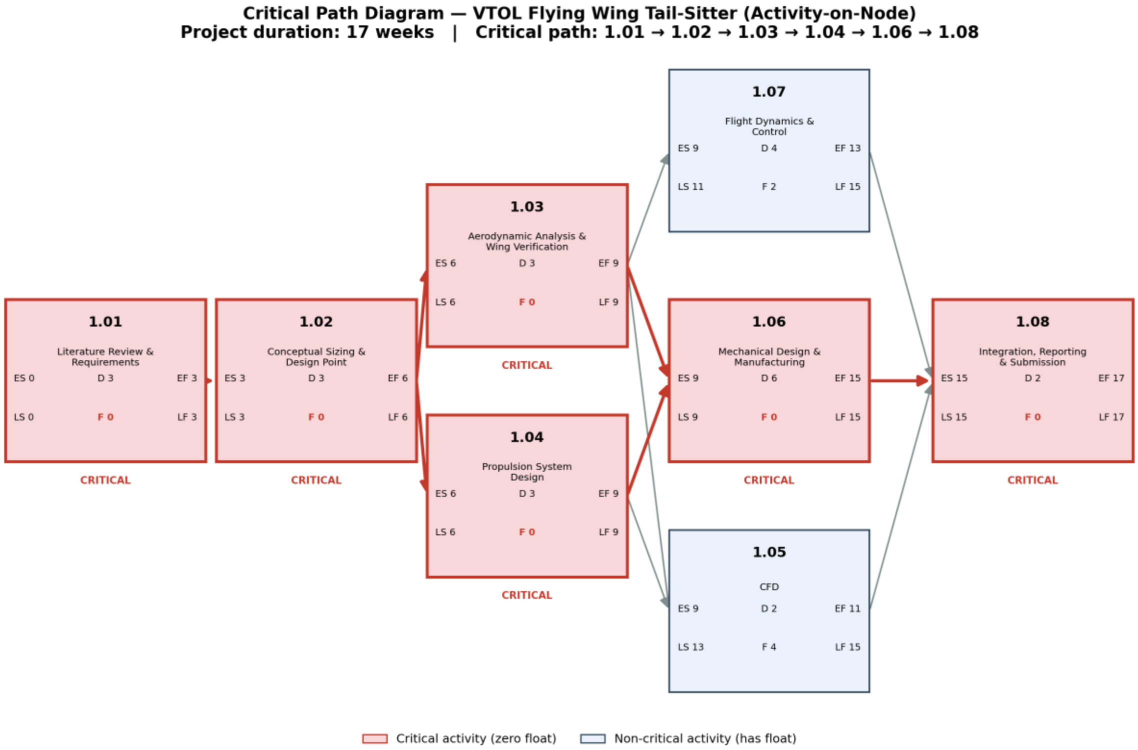


Figure 2 Activity-on-node critical path diagram

3.1 CPM Table

WBS	Phase	Dur	ES	EF	LS	LF	Status
1.01	Literature Review & Requirements	3	0	3	0	3	Critical
1.02	Conceptual Sizing & Design Point	3	3	6	3	6	Critical
1.03	Aerodynamic Analysis & Wing Verification	3	6	9	6	9	Critical
1.04	Propulsion System Design	3	6	9	6	9	Critical
1.05	CFD	2	9	11	13	15	Float 4 wk
1.06	Mechanical Design & Manufacturing	6	9	15	9	15	Critical
1.07	Flight Dynamics & Control	4	9	13	11	15	Float 2 wk
1.08	Integration, Reporting & Submission	2	15	17	15	17	Critical

Critical path: 1.01 → 1.02 → 1.03 / 1.04 (parallel, both critical) → 1.06 → 1.08. Mechanical Design & Manufacturing (WBS 1.06) is the single largest driver of total duration at six weeks and has zero float any slip in structural design, connector printing, or assembly pushes the final submission date directly. CFD (WBS 1.05) and Flight Dynamics & Control (WBS 1.07) carry 4 and 2 weeks of float respectively, so they can absorb minor delays without affecting the finish date, but should not be treated as low-priority their float will be consumed if WBS 1.03/1.04 run late.

4. Risk Assessment

Fourteen risks were identified across the aerodynamic, structural, propulsion, control, and project-management work packages, scored on a 1–5 probability × 1–5 impact scale (score = probability × impact).

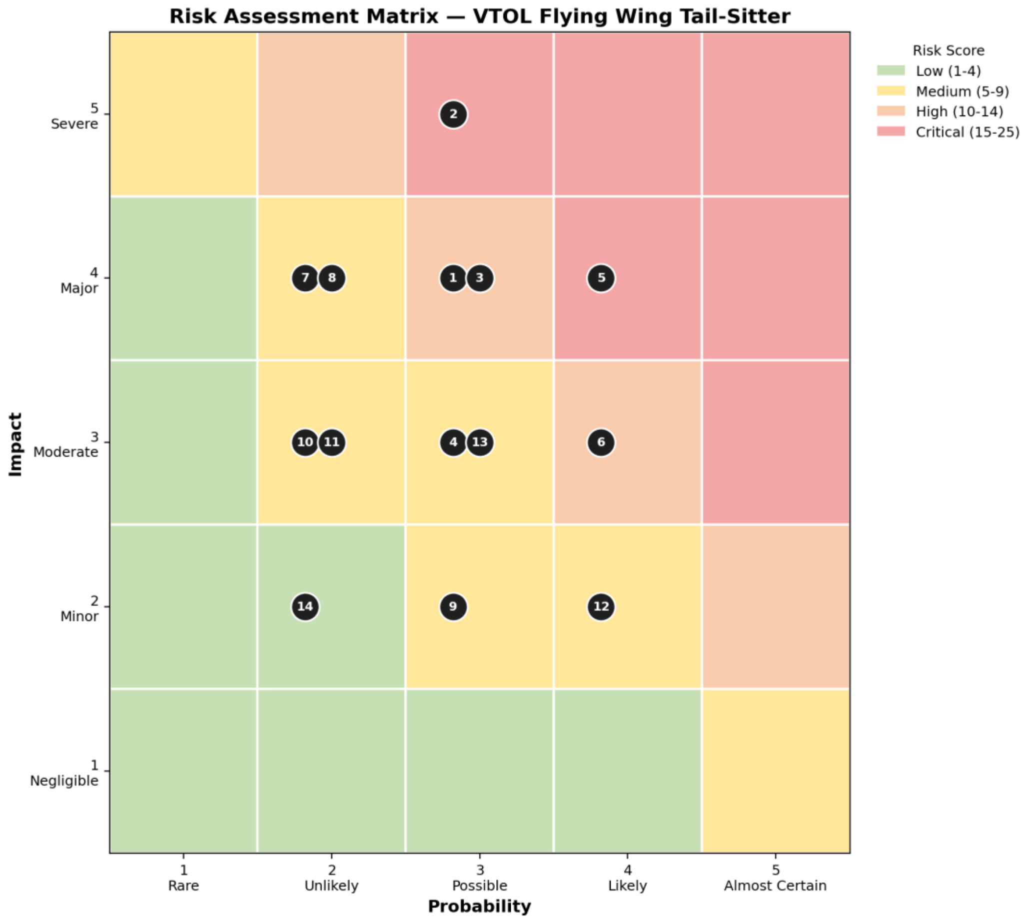


Figure 3 Risk assessment matrix (numbers correspond to Risk IDs in the register below)

4.1 Risk Register

ID	Risk	P	I	Score	Owner	Mitigation
R1	Propulsion candidate fails hover/cruise verification late	3	4	**12**	Propulsion Lead	Carry two backup motor/battery/propeller combinations through hover verification (WBS 1.04.03) instead of a single candidate.
R2	FEA reveals motor mount / spar under-strength, requiring redesign	3	5	**15**	Structures Lead	Run a preliminary motor-mount/spar FEA check as soon as loads are available, before locking final geometry.
R3	3D-printed wing connector fails under load (material/print defect)	3	4	**12**	Structures Lead	Physically load-test one printed connector sample against the FEA-predicted factor of safety before final production.
R4	CFD results diverge from XFLR5 predictions, delaying propulsion sizing	3	3	**9**	Aerodynamics Lead	Cross-check CFD domain/mesh assumptions against XFLR5 inputs early; treat CFD as a refinement, not the sole basis for propulsion sizing.
R5	Weight growth pushes design above matching-plot weight assumption	4	4	**16**	Structures / PM	Track cumulative weight against the WBS 1.06 weight budget continuously; re-run the matching plot if growth exceeds ~5%.
R6	Manufacturing lead time (printing, composite layup) overruns schedule	4	3	**12**	PM / Structures Lead	Order or print long-lead items (motor mount, connectors) immediately after design freeze; retain schedule float within WBS 1.06.
R7	Battery/motor/propeller components unavailable or substituted late	2	4	**8**	Propulsion Lead	Identify two to three substitute components per item during the WBS 1.04.02 database compilation.
R8	Control model instability found late in 3DOF simulation	2	4	**8**	Control Lead	Run preliminary linearized stability checks as soon as stability derivatives are available (parallel with WBS 1.03), not only at final 3DOF sim.

ID	Risk	P	I	Score	Owner	Mitigation
R9	Team member unavailability disrupts cross-discipline coordination	3	2	**6**	Project Manager	Cross-train sub-team members on adjacent WBS elements; maintain shared, up-to-date documentation.
R10	Software/license access (XFLR5, CFD, FEA tools) interruption	2	3	**6**	Project Manager	Confirm tool/license access at the start of each analysis phase; keep local backups of analysis outputs.
R11	Rib spacing / spar sizing errors discovered during assembly	2	3	**6**	Structures Lead	Dry-fit ribs, spars, and connectors before final bonding/assembly.
R12	Report compilation & integration delayed by late sub-team inputs	4	2	**8**	Project Manager	Set internal sub-team documentation deadlines one week ahead of the WBS 1.08.03 compilation start.
R13	Advisor-requested design changes after review	3	3	**9**	Project Manager	Schedule an interim advisor review around week 9 (end of aero/propulsion) rather than only at final submission.
R14	Budget constraint limits material/motor selection options	2	2	**4**	Project Manager	Lock a bill-of-materials budget ceiling early, during WBS 1.06.05 material selection.

4.2 Highest-Priority Risks

R2 (motor mount / spar under-strength found in FEA, score 15) and R5 (weight growth invalidating the matching-plot assumption, score 16) are the two highest-scoring risks and both sit on or feed the critical path (WBS 1.06). They should be checked as early gate reviews rather than left to final verification, since a failure discovered late has no float to absorb it.